

An Exploratory Computational Model of Algorithmic Governance in Platform Economies: Simulating Regulatory Feedback, Endogenous Capture, and Systemic Fragility

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Abstract

This paper presents an exploratory simulation-based framework for analyzing non-linear dynamics in digital platform ecosystems under automated regulatory feedback. We model a multi-sided platform as a coupled dynamical system governed by user capacity, data accumulation, algorithmic optimization, and endogenous governance integrity.

Rather than claiming analytical optimality or equilibrium proofs, the framework focuses on computationally observed behavioral regimes under parameter variation. Regulatory integrity is treated as a dynamic variable subject to degradation via adversarial optimization and metric gaming.

Numerical simulations indicate that increasing regulatory feedback intensity does not monotonically improve system stability. Instead, it can produce three qualitative regimes: (i) monopolistic lock-in, (ii) bounded oscillatory dynamics, and (iii) instability associated with governance degradation under high enforcement intensity.

We employ a surrogate linearization approach using finite-difference Jacobian estimation around empirical time-averaged states to characterize local stability trends. The framework is intended as a computational laboratory for exploring structural risks in rule-based algorithmic governance systems.

1 Introduction

Digital platform ecosystems exhibit strong network effects, high switching costs, and data-driven feedback loops that tend to produce concentrated market structures. Traditional antitrust enforcement operates on slow regulatory timescales compared to the rapid evolution of digital markets.

This mismatch has motivated interest in automated or algorithmic regulatory systems capable of continuous monitoring and intervention. However, coupling such regulatory feedback mechanisms with non-linear platform dynamics may introduce unintended systemic behaviors.

This paper develops an exploratory computational model to investigate these interactions. The goal is not to derive optimal policy rules, but to map qualitative system behaviors and identify potential structural failure modes.

2 Model Ontology

The framework is not intended as a predictive econometric model or a deployable control system. It is a generative computational environment designed to explore non-linear interactions between platform growth dynamics and regulatory feedback mechanisms under bounded assumptions.

3 System Dynamics

We model the system using four coupled state variables:

- $U(t)$: user capacity
- $D(t)$: data accumulation
- $A(t)$: algorithmic dominance
- $G(t)$: governance integrity

3.1 User Capacity

$$\frac{dU}{dt} = \alpha AU \left(1 - \frac{U}{N}\right) - C_s U \quad (1)$$

3.2 Data Accumulation

$$\frac{dD}{dt} = \beta U - \delta \gamma D \quad (2)$$

3.3 Algorithmic Dominance

$$\frac{dA}{dt} = \log(1 + D)(1 - A) - \mu A \quad (3)$$

3.4 Governance Integrity

$$\frac{dG}{dt} = \rho(1 - G) - \omega \frac{U}{N} \gamma G \quad (4)$$

4 Regulatory Feedback Mechanism

Regulatory intervention is modeled as a proportional control function activated above a concentration threshold:

$$\gamma = k \cdot \max\left(0, \frac{U}{N} - \theta\right) \cdot G \quad (5)$$

Switching costs are defined as:

$$C_s = \sigma_0 e^{-1.5\gamma} (2 - G) \quad (6)$$

5 System Calibration

Parameters are chosen to reflect stylized properties of digital platform ecosystems:

Parameter	Interpretation	Value
N	Market capacity	1000
α	Growth rate	0.12
β	Data extraction rate	0.08
δ	Regulatory depletion efficiency	0.15
μ	Algorithmic decay	0.05
σ_0	Baseline switching cost	0.85
θ	Intervention threshold	0.80
ω	Capture intensity	0.04
ρ	Audit recovery rate	0.02

6 Stability Analysis

Because the system does not generally converge to fixed points under moderate to high feedback gain, classical linearization around equilibrium is not applicable.

We instead compute a surrogate Jacobian matrix around the empirical time-averaged state:

$$J_{ij} \approx \frac{f_i(S + \epsilon e_j) - f_i(S)}{\epsilon} \quad (7)$$

Eigenvalues of J are used as local indicators of stability transitions via $\max \Re(\lambda)$.

7 Computational Observations

Simulation sweeps over the regulatory gain parameter k reveal three qualitative regimes:

- **Low feedback:** Monopolistic convergence due to insufficient regulatory strength.
- **Intermediate feedback:** Bounded oscillatory dynamics driven by activation-deactivation cycles of regulation.
- **High feedback:** Instability associated with degradation of governance integrity and increased adversarial optimization.

These regimes should be interpreted as computationally observed patterns rather than analytically proven bifurcation structures.

8 Robustness Considerations

The observed qualitative regimes persist under variations in initial conditions and under stochastic perturbations introduced into the regulatory feedback function. This suggests structural robustness of the behavioral patterns within the modeled parameter space.

9 Discussion

The results indicate that automated regulatory feedback in platform ecosystems may introduce unintended dynamic effects. In particular, overly aggressive feedback mechanisms can incentivize adversarial adaptation by dominant platforms and degrade institutional effectiveness.

The framework highlights the importance of considering endogenous institutional fragility when designing algorithmic governance systems.

10 Conclusion

This paper presents a computational framework for exploring non-linear dynamics in platform economies under algorithmic regulatory feedback. The results suggest that regulatory intensity is not monotonically associated with improved stability and may induce structural fragility under certain conditions.

Rather than proposing an optimal regulatory design, this work maps qualitative behavioral regimes and highlights potential failure modes in automated governance systems.

References

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